

Marishka M. Mehta

Computational Scientist · PhD Candidate in Computational Psychiatry

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Summary

Data scientist and PhD candidate in computational psychiatry with 6+ years of experience in experimental design, Bayesian and machine-learning modeling, and analysis of large longitudinal clinical datasets. Author of 10+ peer-reviewed papers in top psychiatry and neuroscience journals. 2nd place, Hurricane Pitch Competition 2024.

Experience

Graduate Research Assistant — Laureate Institute for Brain Research, Tulsa, OK Aug 2022 – present

Advisor: Dr. Ryan Smith

- Led analysis of an approach-avoidance conflict task across large, longitudinal psychiatric samples; combined frequentist and Bayesian methods with an ensemble classification pipeline that predicts diagnostic group membership out of sample.
- Published a preregistered behavioral study; built reproducible analysis pipelines for behavioral and fMRI data on SLURM-based HPC.

Clinical Research Coordinator — Depression and Anxiety Center, Icahn School of Medicine at Mount Sinai, NY Jul 2021 – Jul 2022

PI: Dr. Laurel Morris

- Designed online cognitive tasks (PsychoPy) and integrated behavioral data with REDCap and mindLAMP2 smartphone-based phenotyping.
 - Built computational and imaging analysis pipelines (MATLAB/VBA toolbox, Python, AFNI, MEICA) for 7T fMRI studies of motivation.
 - Ran IRB submissions, clinical assessments, and participant recruitment for depression and anxiety studies.
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Technical Skills

Programming · Python (NumPy, pandas, scipy, matplotlib, Jupyter), R, MATLAB, Git

Modeling & ML · Bayesian hierarchical modeling (VBA toolbox), reinforcement learning (model-based / model-free), ensemble modeling (caretEnsemble, R), cross-validation

Neuroimaging · 3T and 7T fMRI; fMRIPrep, FSL, SPM, AFNI, MEICA, nilearn

Experimental design & data collection · PsychoPy, jsPsych, REDCap

Reproducibility & open science · Preregistration, reproducible pipelines, SLURM HPC

Selected Work

- **Approach-avoidance diagnostic assessment** — predictive modeling pipeline that classifies psychiatric diagnoses out of sample; interpretable markers for clinical assessment, prognosis, and treatment response. *Computational Psychiatry*, 2025. [Link](#).
 - **Digital phenotyping of mental health** — smartphone-based activity and motivation measures that track depression and anxiety in the real world. *npj Digital Medicine*, 2025. [Link](#).
 - **Motivation and agency as symptom-dimension markers** — computational parameters linked to anhedonia, stress, and anxiety; subjective agency over rewards reduced in depression and anxiety. *Psychological Medicine*, 2024; *PLOS ONE*, 2023.
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Education

- **PhD, Data Science & Psychology** — The University of Tulsa · expected 2027
 - **M.A., Psychology** — New York University · 2021
 - **M.Sc., Neuroscience (Neurasmus Dual Diploma)** — University of Bordeaux & Laval University · 2019
 - **B.Sc. (Hons.), Biomedical Science** — Queen Mary, University of London · 2015
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Selected Awards

- **Dissertation Completion Fellowship** — University of Tulsa · 2026–2027
- **2nd Place, Hurricane Pitch Competition (\$3,500)** — University of Tulsa · 2024
- **Trainee Professional Development Award (\$1,000)** — Society for Neuroscience · 2025